

Xiaohan Zhang

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RESEARCH SUMMARY

Robot Learning and Planning

My research goal is to build robots that generalize across diverse, long-horizon tasks in complex household environments. Toward this goal, I explore robot foundation models, structured world representations, and task and motion planning (TAMP).

INDUSTRIAL EXPERIENCES

Boston Dynamics AI Institute

Robotist

Large-scale sim-to-real generalist policy learning

Cambridge, MA

May 2024 – Present

Meta, FAIR Labs

Research Scientist Intern

Vision-language models and 3D scene representations for task planning, embodied question answering, and primitive skill learning

Pittsburgh, PA

May 2023 – Dec. 2023

Google DeepMind, Robotics

Student Researcher

Task and motion planning for whole-body manipulation

Mountain View, CA

Mar. 2022 – Aug. 2022

ACADEMIC BACKGROUND

State University of New York at Binghamton

Ph.D. in Computer Science

M.S. in Computer Science

Thesis: Symbol Grounding for Task and Motion Planning in Robotics

Binghamton, NY

Aug. 2020 – May 2024

Aug. 2019 – May 2020

University of Texas at Austin

Visiting Ph.D. Student

Austin, TX

Jan. 2023 – Apr. 2023

Renmin University of China

B.S. in Computer Science

Beijing, China

Sep. 2015 – May 2019

SELECTED PUBLICATIONS

See my [google scholar](#) for a full list including pre-prints.
(*indicates equal contributions)

JOURNAL ARTICLES

- [J1] Ivan Kapelyukh, **Xiaohan Zhang**, Stephen James, Laura Herlant, and Edward Johns.
ZeroBot: Learning from Scratch in Minutes with Generative Real2Sim.
IEEE Robotics and Automation Letters (RA-L), 2026
- [J2] **Xiaohan Zhang***, Yan Ding*, Yohei Hayamizu*, Zainab Altaweel*, Yifeng Zhu, Yuke Zhu, Peter Stone, Chris Paxton, and Shiqi Zhang.
[LLM-GROP: Visually Grounded Robot Task and Motion Planning with Large Language Models.](#)
The International Journal of Robotics Research (IJRR), 2025
- [J3] **Xiaohan Zhang**, Saeid Amiri, Jivko Sinapov, Jesse Thomason, Peter Stone, and Shiqi Zhang.

Multimodal Embodied Attribute Learning by Robots for Object-Centric Action Policies.

Autonomous Robots, 2023

- [J4] Yan Ding, **Xiaohan Zhang**, Saeid Amiri, Nieqing Cao, Hao Yang, Chad Esselink, and Shiqi Zhang.

[Integrating Action Knowledge and LLMs for Task Planning and Situation Handling in Open Worlds.](#)

Autonomous Robots, 2023

- [J5] Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang.

[Learning to Ground Objects for Robot Task and Motion Planning.](#)

IEEE Robotics and Automation Letters (RA-L), 2022

CONFERENCE PAPERS

- [C1] Suning Huang, Qianzhong Chen, **Xiaohan Zhang**, Jiankai Sun, and Mac Schwager. [ParticleFormer: A 3D Point Cloud World Model for Multi-Object, Multi-Material Robotic Manipulation.](#)

Conference on Robot Learning (CoRL), 2025

- [C2] Arjun Majumdar*, Anurag Ajay*, **Xiaohan Zhang***, Pranav Putta, Sriram Yendamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Sasha Sax, and Aravind Rajeswaran.

[OpenEQA: Embodied Question Answering in the Era of Foundation Models.](#)

The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024

- [C3] Priyam Parashar, Vidhi Jain, **Xiaohan Zhang**, Jay Vakil, Sam Powers, Yonatan Bisk, and Chris Paxton.

[SLAP: Spatial-Language Attention Policies.](#)

Conference on Robot Learning (CoRL), 2023

- [C4] **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuqian Jiang, Yuke Zhu, Peter Stone, and Shiqi Zhang.

[Symbolic State Space Optimization for Long Horizon Mobile Manipulation Planning.](#)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023

- [C5] Yan Ding*, **Xiaohan Zhang***, Chris Paxton, and Shiqi Zhang.

[Task and Motion Planning with Large Language Models for Object Rearrangement.](#)

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023

- [C6] Thomas Lew, Sumeet Singh, Mario Prats, Jeffrey Bingham, Jonathan Weisz, Benjie Holson, **Xiaohan Zhang**, Vikas Sindhwani, Yao Lu, Fei Xia, Peng Xu, Tingnan Zhang, Jie Tan, and Montserrat Gonzalez.

[Robotic Table Wiping via Reinforcement Learning and Whole-body Trajectory Optimization.](#)

IEEE International Conference on Robotics and Automation (ICRA), 2023

- [C7] **Xiaohan Zhang**, Yifeng Zhu, Yan Ding, Yuke Zhu, Peter Stone, and Shiqi Zhang. [Visually Grounded Task and Motion Planning for Mobile Manipulation.](#)

IEEE International Conference on Robotics and Automation (ICRA), 2022

- [C8] **Xiaohan Zhang**, Jivko Sinapov, and Shiqi Zhang.

[Planning Multimodal Exploratory Actions for Online Robot Attribute Learning.](#)

Robotics: Science and Systems (RSS), 2021

- [C9] Kishan Chandan, Jack Albertson, **Xiaohan Zhang**, Xiaoyang Zhang, Yao Liu, and Shiqi Zhang.
[Learning to Guide Human Attention on Mobile Telepresence Robots with 360 Vision.](#)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021
- [C10] Yan Ding, **Xiaohan Zhang**, Xingyue Zhan, and Shiqi Zhang.
[Task-Motion Planning for Safe and Efficient Urban Driving.](#)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020

HONORS & AWARDS

- **Outstanding Reviewer for IEEE Robotics and Automation Letters, 2024**
- **SUNY Binghamton Excellence in Research in Computer Science, 2024**

MENTORSHIP

- **Zixing Wang:** PhD student at Purdue University.
- **Zifan Xu:** PhD student at UT Austin.
- **Rutav Shah:** PhD student at UT Austin.
- **Ivan Kapelyukh:** Research engineer at Google DeepMind; PhD student at Imperial College London.

ACADEMIC SERVICES

- **Organizer:** ICRA Competition on What Bimanuals Can Do (WBCD), 2026
- **Organizer:** CVPR Workshop on Foundation Models Meet Embodied Agents, 2025
- **Reviewer:** IJRR, RA-L, ICRA, IROS, CoRL, AAAI, IJCAI

TECHNICAL SKILLS

- **Programming Languages:** Python (proficient), C/C++
- **Robotics Software and Simulators:** IsaacLab, ROS, Habitat
- **Deep Learning Frameworks:** PyTorch, Ray